GCP vs Azure

Data Engineering Service Mapping

Complete Platform Comparison

1. CORE DATA SERVICES COMPARISON

Category	GCP	Azure	Notes
Object Storage	Google Cloud Storage (GCS)	Azure Blob Storage Azure Data Lake Storage Gen2 (ADLS)	ADLS = Blob + hierarchical namespace
Data Warehouse	BigQuery	Azure Synapse Analytics	Synapse = integrated analytics platform
Batch Spark	Dataproc	Azure Databricks HDInsight Synapse Spark	Multiple options in Azure
Streaming	Dataflow (Apache Beam)	Azure Stream Analytics Event Hubs	Different paradigms
Orchestration	Cloud Composer (Airflow)	Azure Data Factory Synapse Pipelines	ADF = visual ETL Composer = code-first
Catalog	Dataplex	Azure Purview	Both: governance + lineage

2. STORAGE COMPARISON

Feature	GCS	Azure Blob Storage	ADLS Gen2
Type	Object storage	Object storage	Object + hierarchical
Hierarchy	Flat (prefix simulation)	Flat (prefix simulation)	True folders/directories
Storage Tiers	4 tiers: • Standard • Nearline • Coldline • Archive	4 tiers: • Hot • Cool • Archive • Premium	Same as Blob Storage + Big Data optimized
Max Object Size	5 TB	190 TB (block blob)	5 TB per file (unlimited append)
Use Case	General object storage	General object storage	Big data analytics (optimized for Hadoop)
Hadoop Compatible	Yes (via connector)	Yes (via connector)	Native (HDFS-compatible)

Key Differences: • **ADLS Gen2** is the recommended storage for big data in Azure (hierarchical namespace + performance optimizations) • **GCS** uses flat namespace with prefix simulation (like AWS S3) • **ADLS Gen2** has native ACLs (POSIX-like) on folders/files, GCS uses IAM • Both support lifecycle policies, versioning, encryption

3. DATA WAREHOUSE COMPARISON

Feature	BigQuery	Azure Synapse Analytics
Architecture	Serverless, columnar MPP	Dedicated SQL pool (provisioned) or Serverless SQL pool
Pricing	Pay per TB scanned	Dedicated: Pay for DWUs 24/7 Serverless: Pay per TB processed
Scaling	Automatic, instant	Dedicated: Manual resize Serverless: Auto
Storage	Decoupled (managed)	Decoupled (ADLS Gen2)
Query Language	Standard SQL	T-SQL (SQL Server dialect)
Streaming Inserts	Native (streaming buffer)	Via Stream Analytics
Time Travel	7 days (up to 90 days)	Not built-in (manual)
Data Loading	Batch (load jobs) Streaming API	COPY command PolyBase (external tables)
Max Query Size	No practical limit	Depends on DWU/memory

Key Insights: • **BigQuery** is fully serverless - zero management, instant scaling • **Synapse Dedicated Pool** is like Redshift - provisioned, requires tuning (distribution, indexes) • **Synapse Serverless** is like BigQuery - pay per query, good for ad-hoc analytics • **Synapse** is an integrated platform (DW + Spark + Pipelines) vs BigQuery (DW only)

4. BATCH PROCESSING (Spark)

Feature	Dataproc	HDInsight	Synapse Spark	Databricks on Azure
What	Managed Hadoop/Spark	Managed Hadoop/Spark	Integrated Spark in Synapse	Unified analytics platform
Management	Managed clusters	Managed clusters	Serverless or provisioned	Managed clusters (optimized)
Startup Time	~5 min (cluster) ~30 sec (serverless)	~10 min	~1 min	~30 sec
Optimization	Manual Spark tuning	Manual Spark tuning	Some auto-tuning	Auto-optimization (Photon)
Integration	GCP services	Azure services	Native Synapse integration	Standalone platform
Use Case	Full Spark control, Hadoop ecosystem	Windows-based Hadoop	Integrated with Synapse DW	Modern data engineering

Recommendations: • **Dataproc** ≈ **HDInsight:** Similar managed Hadoop/Spark (HDInsight has Windows option) • **Synapse Spark:** Use if already in Synapse ecosystem, want tight DW integration • **Databricks:** Best developer experience, fastest, most features (but highest cost)

5. ORCHESTRATION & ETL

Feature	Cloud Composer	Azure Data Factory (ADF)	Synapse Pipelines
Type	Code-first (Python)	Visual + code	Visual + code (same as ADF)
Engine	Apache Airflow	Proprietary	Proprietary
DAG Definition	Python code	JSON (ARM templates)	JSON
Activities	Airflow operators	80+ activities (Copy, Dataflow, etc.)	Same as ADF
Scheduling	Cron, sensors	Triggers (schedule, tumbling, event)	Same as ADF
Monitoring	Airflow UI	ADF monitor + Log Analytics	Synapse Studio
Complexity	Code (more flexible)	GUI (easier for simple)	GUI (easier)
Best For	Complex workflows, code-first teams	Visual ETL, Azure-native	Synapse-integrated workflows

Key Differences: • **Cloud Composer (Airflow):** Code-first, maximum flexibility, steeper learning curve • **Azure Data Factory:** Visual drag-and-drop, easier for simple ETL, less flexible for complex logic • **ADF** has built-in connectors to 90+ data sources (including on-prem via Integration Runtime) • You can run Airflow on Azure using AKS (Kubernetes) or Azure Container Instances

6. STREAMING DATA

Feature	GCP Pub/Sub + Dataflow	Azure Event Hubs + Stream Analytics	Azure Event Hubs + Spark Streaming
Messaging	Pub/Sub (global pub-sub)	Event Hubs (Kafka-compatible)	Event Hubs
Processing	Dataflow (Apache Beam)	Stream Analytics (SQL-based)	Synapse/Databricks Spark
Language	Java, Python, Go	SQL (declarative)	Python, Scala
Windowing	Beam windows	Tumbling, hopping, sliding, session	Spark windows
State	Beam state API	Built-in	Spark state
Scalability	Auto-scaling	Streaming Units (manual)	Auto-scaling
Exactly-once	Yes	Yes (with checkpointing)	Yes
Use Case	Complex streaming logic	Simple SQL transformations	Complex Spark-based streaming

Equivalents: • **Pub/Sub** ≈ **Event Hubs:** Similar messaging platforms (Event Hubs is Kafka-compatible) • **Dataflow** ≈ **Stream Analytics:** Different paradigms (Beam vs SQL) • **Dataflow** ≈ **Spark Streaming:** More similar (both support complex logic)

7. SERVERLESS COMPUTE

Feature	Cloud Functions	Azure Functions
Trigger Types	HTTP, Pub/Sub, Storage, Firestore, etc.	HTTP, Timer, Queue, Blob, Event Hub, etc.
Languages	Python, Node.js, Go, Java, .NET, Ruby, PHP	C#, Python, JavaScript, Java, PowerShell, TypeScript
Timeout	9 min (HTTP) 60 min (background)	5 min (Consumption) 30 min (Premium)
Cold Start	Yes (~1-2 sec)	Yes (~1-2 sec)
Pricing	Invocations + GB-sec	Invocations + GB-sec

Nearly Identical: Both serve same purpose - event-driven, lightweight compute.

8. DATA GOVERNANCE & CATALOG

Feature	Dataplex	Azure Purview
Purpose	Data governance, catalog, quality, lineage	Unified data governance, catalog, compliance
Discovery	Auto-discovery from GCS, BigQuery, Cloud SQL	Auto-discovery from ADLS, SQL, Synapse, Power BI
Lineage	Data lineage tracking	End-to-end lineage (data + reports)
Classification	Auto-classification (PII detection)	Auto-classification (100+ built-in types)
Access Control	IAM integration	RBAC + integration with Azure AD
Business Glossary	Yes	Yes
Data Quality	DQ rules + monitoring	Data quality not native (use ADF/Synapse)
Cross-cloud	GCP only	Azure + AWS + GCP (multi-cloud)

Key Difference: • **Azure Purview** supports multi-cloud governance (Azure + AWS + GCP) • **Dataplex** is GCP-only but has tighter integration with GCP services • Both provide: data discovery, classification, lineage, business glossary

9. COMPLETE SERVICE MAPPING

GCP Service	Azure Equivalent	Notes
GCS	ADLS Gen2	Azure: hierarchical namespace
BigQuery	Synapse Dedicated Pool Synapse Serverless	Synapse: integrated platform
Dataproc	HDInsight Synapse Spark	Multiple options in Azure
Dataflow	Stream Analytics Dataflow (preview)	Azure Dataflow coming
Cloud Composer	Data Factory Airflow on AKS	ADF: GUI, Airflow: code
Pub/Sub	Event Hubs Service Bus	Event Hubs: Kafka-compatible
Cloud Functions	Azure Functions	Nearly identical
Cloud Run	Container Apps	Serverless containers
GKE	AKS	Managed Kubernetes
Dataplex	Purview	Purview: multi-cloud
BigQuery ML	Synapse ML Azure ML	Multiple ML options
Looker	Power BI	Azure: tighter integration
IAM	Azure AD + RBAC	Similar concepts
Secret Manager	Key Vault	Identical purpose
Cloud Logging	Log Analytics Monitor	Azure: more services

10. INTERVIEW TRANSLATION GUIDE

They Ask	You Respond
Experience with ADLS Gen2?	I've worked extensively with GCS which is similar object storage. ADLS Gen2 adds hierarchical nam
Synapse Analytics experience?	I'm coming from BigQuery which is serverless. Synapse offers both dedicated pools (like Redshift) a
Azure Data Factory vs Composer?	I've used Cloud Composer (managed Airflow) extensively. ADF is more visual/GUI-based which is g
Event Hubs vs Pub/Sub?	Event Hubs is Kafka-compatible messaging, similar to Pub/Sub's pub-sub pattern. Both provide reli

Key Translation Points:

- Your **GCS experience** → ADLS Gen2 (add hierarchical namespace concept)
- Your **BigQuery expertise** → Synapse Serverless or Dedicated Pool
- Your **Dataproc jobs** → HDInsight or Synapse Spark (Spark code mostly portable)
- Your **Cloud Composer DAGs** → Azure Data Factory or Airflow on AKS
- Your **Pub/Sub streaming** → Event Hubs + Stream Analytics

11. KEY PLATFORM DIFFERENCES

Philosophy:

• **GCP:** Best-of-breed individual services, serverless-first, developer-friendly • **Azure:** Integrated platforms (Synapse = DW + Spark + Pipelines), enterprise-focused, on-prem integration

Data Warehouse:

• **BigQuery:** Fully serverless, zero management, instant scaling • **Synapse:** Choice of dedicated (manual tuning) or serverless (BigQuery-like)

Orchestration:

• **Cloud Composer:** Code-first (Python), maximum flexibility • **Azure Data Factory:** GUI-first, easier for simple ETL, less flexible

Spark:

• **GCP:** Dataproc (managed clusters) or Dataproc Serverless • **Azure:** HDInsight, Synapse Spark, OR Databricks (recommended for modern workloads)

Integration:

• **Azure:** Strong on-prem integration (Azure Arc, hybrid scenarios) • **GCP:** Cloud-native focus, less on-prem tooling

Summary:

Your GCP experience translates well to Azure. The core data engineering concepts (distributed computing, data lakes, partitioning, orchestration) are identical. Azure tends to offer more integrated platforms (Synapse) vs GCP's best-of-breed services. Both clouds are converging - Azure is adding more serverless options, GCP is adding integrated platforms.